THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** heat is used to change the state	Latent heat idea.
2	**(c)** 100 g	Mass is conserved.
3	**(d)** it changed to vapour and mixed with air	Matter is conserved.
4	**(d)** melting and boiling	States change.
5	**(a)** 10 mL	50/5.
6	**(a)** smaller volume and large surface to depth ratio	Less water.
7	**(c)** the colour of the container	Not relevant.
8	**(c)** 60 L	2 x 30.
9	**(c)** melting of ice	Water can be refrozen.
10	**(a)** increase	Larger area.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vapour cannot escape (1); Air in the bag becomes saturated (1).	1+1
12	Air's vapour touches the cold surface (1); It condenses on the outside (1).	1+1
13	Fall (1); Wells may dry (1).	1+1
14	Pressure inside is higher (1); Higher pressure raises the boiling point (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	4 x 30 = 120 L (1); 120/2000 (1); 6 per cent (1).	1+1+1
16	Moist air rises over hills (1); It cools and condenses (1); Clouds drop rain (1).	1+1+1
17	More evaporation (1); Irregular rain (1); Melting glaciers (1).	1+1+1
	OR: Cover a bowl and an open bowl (1); Open bowl loses level (1); Covered shows droplets on the lid (1).	1+1+1
18	Temperature is below 0 °C (1); Snow is compact (1); Insulates inside (1).	1+1+1
	OR: Forms a barrier (1); Stops vapour escaping (1); Saves water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Identical wet cloths (1); Place in three locations (1); Weigh at intervals (1); Greatest loss shows the fastest drying (1); Conclusion with reasons (1).	1+1+1+1+1
	OR: Ice warms to 0 °C (1); Flat at 0 °C: melting (1); Water warms to 100 °C (1); Flat at 100 °C: boiling (1); Heat used for change of state (1).	1+1+1+1+1
20	Sea water in a basin (1); Sun heats, evaporation (1); Vapour condenses on the cool cover (1); Drops collect in a container (1); Salt left behind (1).	1+1+1+1+1
	OR: Taps at home (1); Reuse water (1); Industrial recycling (1); Dams and canals (1); Policies and awareness (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 180 mL	3600/20.
2	**(c)** it increases	More evaporation.
3	**(c)** evaporation and condensation	Local water cycle.
4	**(d)** 50 L	0.05 m³ = 50 L.
5	**(c)** transpiration part of the cycle	Water vapour from leaves.
6	**(d)** steam releases heat on condensing	Condensation releases heat.
7	**(d)** salts brought by rivers and left after evaporation	Accumulation.
8	**(d)** more transpiration adds moisture	Water vapour to the air.
9	**(c)** evaporation	Heat.
10	**(d)** evaporation and condensation	Distillation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Only vapour rises (1); Salt is left behind (1).	1+1
12	Evaporation balances (1); Water returns to the cycle (1).	1+1
13	Ice occupies more volume (1); Expansion on freezing (1).	1+1
14	Evaporation takes heat (1); Cooling effect (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	86400 drops (1); 86400 / 20 = 4320 mL (1); 4.32 L (1).	1+1+1
16	Air temperature below 0 °C (1); Vapour changes directly to ice crystals (1); Thin white ice layer (1).	1+1+1
17	Heat (1); Moving air (1); Tumbling increases exposure (1).	1+1+1
	OR: Pressure builds (1); Steam expands (1); Vessel bursts (1).	1+1+1
18	Mulching (1); Planting trees as windbreaks (1); Irrigating in the evening (1).	1+1+1
	OR: Water is removed by drying (1); Preserves food (1); Sun used (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sea gives vapour (1); Forests add vapour (1); Mountains trigger rain (1); Rivers carry water back (1); Pollution and deforestation disturb (1).	1+1+1+1+1
	OR: Waste dumping (1); Sewage (1); Treatment plants (1); Awareness (1); Laws (1).	1+1+1+1+1
20	Uneven rain (1); Overuse of groundwater (1); Harvest rain (1); Efficient irrigation (1); Protect ponds (1).	1+1+1+1+1
	OR: Surface: rivers and lakes (1); Groundwater: wells (1); Surface prone to pollution (1); Groundwater limited (1); Both need care (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **A Journey through States of Water** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** building percolation pits	Allows seepage.
2	**(d)** snow melts	Glaciers feed rivers.
3	**(d)** condensation	Cooled vapour.
4	**(d)** less evaporation and rain	Less energy.
5	**(d)** burning of wood	Not water movement.
6	**(a)** evaporation from pores takes away heat	Evaporative cooling.
7	**(a)** decreases	Ice is less dense.
8	**(a)** ice occupies more volume than water	Expands on freezing.
9	**(c)** glaciers and ice caps	Frozen stores.
10	**(d)** stays cool by evaporation	Cool drinking water.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They store water as ice (1); Melt in summer to feed rivers (1).	1+1
12	Air near the ground cools (1); Vapour condenses into tiny droplets (1).	1+1
13	They affect rivers and people (1); Need safe design (1).	1+1
14	Water on the skin evaporates (1); Takes away heat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Winds move clouds (1); Clouds cool and rain (1); Rain falls far from the source (1).	1+1+1
16	Milk forms a layer of cream (1); Traps steam (1); Bubbles push it up (1).	1+1+1
17	Flood: water overflows land (1); Shower: short rain (1); Drainage determines the result (1).	1+1+1
	OR: Sea water in shallow pans (1); Sun evaporates water (1); Salt crystals remain (1).	1+1+1
18	Water on skin evaporates (1); Evaporation needs heat (1); Heat is taken from skin (1).	1+1+1
	OR: Fast evaporation (1); Water wasted (1); Leaves may burn (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Closed jar: no loss of vapour (1); Water evaporated and condensed (1); Mass stays the same (1); Matter conserved (1); Local water cycle (1).	1+1+1+1+1
	OR: Farming: crop planning (1); Floods: drainage (1); City: storm water (1); Harvesting (1); Weather forecasts (1).	1+1+1+1+1
20	Ice cube to water (1); Water to steam on a stove (1); Steam to droplets on a lid (1); Water to ice in a freezer (1); Same substance, different states (1).	1+1+1+1+1
	OR: Cylinder with funnel (1); Measure depth daily (1); Record monthly total (1); Compare months (1); Use for harvesting plan (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.